02/27/23 Signals

I Minimum Mean Square Error

$$f[t] = [f[t] \quad f[t-1] \quad f[t-2] \cdots f[t-m]]$$

$$h = [h[0] \quad [h_1] \cdots h[m]]$$

Let

$$y[t] = h^T f[t]$$

$$e[t] = d[t] - y[t]$$

Desired signal

Actual signal output

We want $d[t] - y[t] \perp$ "data" such that we can minimize $E[(e[t])^2]$. Thus to minimize the error we know we want the error $e[t]$ to be orthogonal to the data, so we write

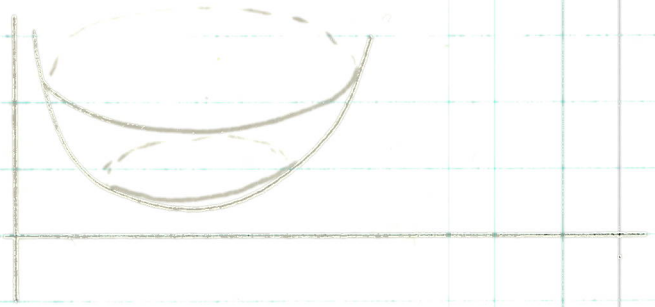
$$E[(d[t] - y[t]) f[t]] = 0$$

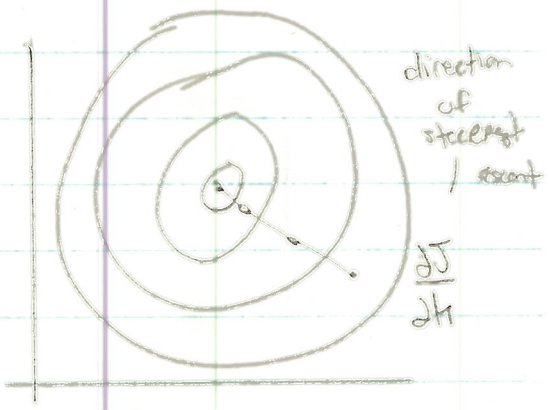
$$\underbrace{E[d[t] f[t]]}_{P} = f[t] f[t]^T h$$

Given the new names we have specified, we write

$$P = Rh$$

II Gradient Descent





$$J(h) = E[(e[t])^2] = E[(d(t) - h^T f[t])^2]$$

$$= E[d(t)^2 - 2d(t)h^T f[t] + h^T f[t] f[t]^T h]$$

$$= E[d[t]^2] - 2h^T E[d[t] f[t]] + h^T E[f[t] f[t]^T] h$$

$$\frac{\partial J}{\partial h} = -2E[d[t] f[t]] + 2E[f[t] f[t]^T] h$$

The Gradient Descent formula is

$$h^{[k+1]} = h^{[k]} - \mu[2E[f[t] f[t]^T] h^{[k]} - 2E[d[t] f[t]]]$$

$$H^{[k+1]} = H^{[k]} - 2\mu \left[E\left[f[t]\, y[t] - d[t]\, f[t] \right] \right]$$

$$= H^{[k]} - \xi \left[E\left[y(t) - d(t) \right) f[t] \right]$$

$$= H^{[k]} + \xi\, E\left[e(t)\, f[t] \right]$$

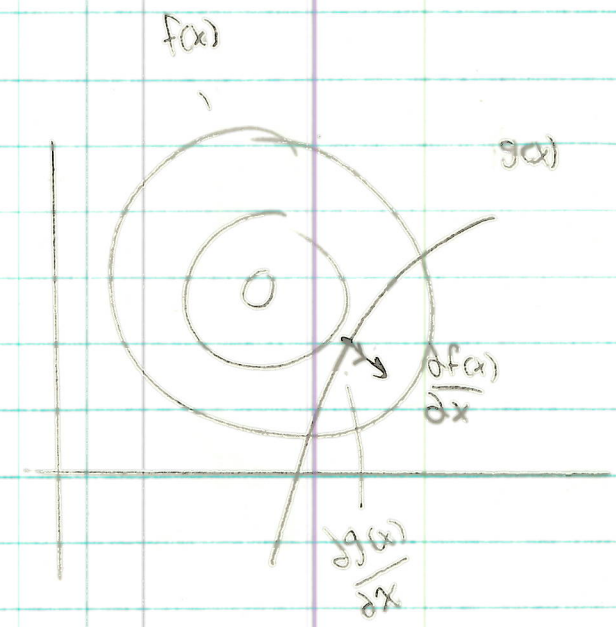
Because we have no real way of taking the expected value, we are just going to throw it away. This generates the stochastic gradient descent.

$$\underbrace{H^{[k+1]} = H^{[k]} + \xi\, e(t)\, f[t]}_{\text{LMS Adaptive Filter}} \qquad \text{where} \quad \xi = 2\mu$$

III Constrained Optimization

$$\min \; f(x)$$

$$\text{s.t.} \quad g(x) = 0$$



Thus

$$\frac{\partial f(x)}{\partial x} - \lambda \frac{\partial g(x)}{\partial x} = 0$$

A) Lagrangian

$$\mathcal{L}(x,\lambda) = f(x) - \lambda(g(x))$$

1) $$\underbrace{\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial f(x)}{\partial x} - \lambda \frac{\partial g(x)}{\partial x} = 0}_{\text{Condition of optimality}}$$

2) $$\frac{\partial \mathcal{L}}{\partial \lambda} = g(x)$$

B) Example

$\min \; x^T A x$ A is symmetric

s.t $x^T x = 1$

$$\mathcal{L}(x,\lambda) = x^T A x - \lambda x^T x$$

$$\frac{\partial \mathcal{L}}{\partial x} = 2Ax - 2\lambda x = 0$$

$\Rightarrow$ $2Ax = 2\lambda x \Rightarrow Ax = \lambda x$

x: eigenvector

λ: eigenvalue

$x^T A x = \lambda x^T x$ Multiply both sides by x^T